

The General Theory of the Construction, Usage, Maintenance, and Deconstruction of Theoretical Models

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Abstract

This paper develops a general theory of theoretical models. The framework treats models as information-processing structures that transform observations into conclusions under finite cognitive, computational, institutional, and physical constraints. We unify perspectives from mathematics, logic, physics, chemistry, biology, economics, finance, psychology, psychiatry, philosophy, political science, international relations, warfare studies, statistics, computer science, data science, and artificial intelligence. We develop a life-cycle theory of models encompassing construction, usage, maintenance, deconstruction, competition, decay, replacement, and meta-theoretical recursion. The resulting framework provides a common language for understanding scientific theories, engineering systems, machine learning models, strategic doctrines, legal frameworks, and social institutions.

The paper ends with “The End”

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1 Introduction

Human beings never interact directly with reality in its entirety. Every observation, measurement, theory, institution, ideology, scientific framework, military doctrine, or machine learning architecture constitutes a model.

The central thesis of this paper is that all finite agents require models because reality contains more information than any finite agent can process. Consequently, civilization itself may be interpreted as a collection of interacting models.

The construction, usage, maintenance, and deconstruction of models therefore become fundamental problems spanning all intellectual disciplines.

2 The Nature of Models

Definition 1 (Theoretical Model). *A theoretical model is a tuple*

$$\mathcal{M} = (\mathcal{O}, \mathcal{S}, \mathcal{P}, \mathcal{A}, \mathcal{D}, \mathcal{C})$$

where

- $\mathcal{O}$ denotes observations,
- $\mathcal{S}$ denotes state variables,
- $\mathcal{P}$ denotes parameters,
- $\mathcal{A}$ denotes assumptions,
- $\mathcal{D}$ denotes dynamical laws,
- $\mathcal{C}$ denotes conclusions.

The model performs the transformation

$$\mathcal{C} = F(\mathcal{O}; \mathcal{S}, \mathcal{P}, \mathcal{A}, \mathcal{D}).$$

2.1 The Universal Modelling Process

Every model may be represented as

$$\text{Observation} \rightarrow \text{Compression} \rightarrow \text{Prediction} \rightarrow \text{Decision} \rightarrow \text{Revision}.$$

3 The Necessity of Models

Definition 2 (Kolmogorov Complexity). *Let*

$$K(\Omega)$$

denote the informational complexity of reality.

Theorem 1 (Information Compression Necessity). *Suppose reality possesses complexity*

$$K(\Omega)$$

and an agent possesses capacity

$$K_A.$$

If

$$K(\Omega) > K_A,$$

then direct reasoning about reality is impossible.

Proof. The agent cannot store, represent, or manipulate all information contained in reality.

Therefore the agent must construct a reduced representation

$$\Omega' \subseteq \Omega.$$

This reduced representation is a model.

□

Corollary 1. *Every finite intelligence necessarily uses models.*

4 Model Construction

4.1 Purpose

No model exists independently of purpose.

Define utility

$$U(M).$$

Model construction solves

$$\max_M U(M)$$

subject to

$$C(M) \leq B,$$

where

- $C(M)$ denotes modelling cost,
- B denotes available resources.

4.2 The Four-Layer Construction Architecture

1. Ontology.
2. State Variables.
3. Dynamical Laws.
4. Objectives.

Discipline	State Variables	Objective
Physics	Position, Velocity	Prediction
Chemistry	Concentrations	Equilibrium
Biology	Population Levels	Survival
Economics	Prices, Output	Welfare
Finance	Returns, Risk	Utility
Political Science	Power, Influence	Stability
Military Science	Force Levels	Victory
Machine Learning	Parameters	Loss Minimization

Table 1: Examples of modelling structures across disciplines.

5 Logic and Epistemology of Models

A model induces a logical system

$$\mathcal{L} = (\mathcal{A}, \mathcal{R}),$$

where

- $\mathcal{A}$ are axioms,
- $\mathcal{R}$ are inference rules.

The conclusions of a model satisfy

$$\mathcal{A} \vdash \mathcal{C}.$$

Thus every model is fundamentally a constrained logical machine.

6 Models in Physics

Physical theories are models describing state evolution.

Newtonian mechanics:

$$F = ma.$$

Relativity:

$$G_{\mu\nu} = 8\pi T_{\mu\nu}.$$

Quantum mechanics:

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi.$$

Each theory provides a compressed representation of reality.

7 Models in Chemistry

Chemical models describe molecular interactions.

Reaction dynamics:



Mass action kinetics:

$$\frac{dC}{dt} = kAB.$$

Chemical models transform microscopic interactions into macroscopic predictions.

8 Models in Biology

Population dynamics:

$$\frac{dN}{dt} = rN \left(1 - \frac{N}{K} \right).$$

Evolutionary fitness:

$$\dot{s}_i = s_i(f_i - \bar{f}).$$

Biological systems continuously generate and update internal models of their environments.

9 Models in Economics and Finance

Economic models simplify decision-making under scarcity.

Consumer optimization:

$$\max U(x)$$

subject to

$$px \leq I.$$

Asset pricing:

$$P_t = E_t [m_{t+1} X_{t+1}].$$

Black-Scholes:

$$dS_t = \mu S_t dt + \sigma S_t dW_t.$$

10 Models in Psychology and Psychiatry

Human cognition relies on internal predictive models.

Prediction error:

$$\epsilon_t = y_t - \hat{y}_t.$$

Belief updating:

$$P(H|D) = \frac{P(D|H)P(H)}{P(D)}.$$

Psychiatric disorders may be interpreted as failures of cognitive modelling systems.

11 Models in Political Science and International Relations

Political systems are collections of interacting strategic models.

Power balance:

$$P_i = M_i + E_i + T_i,$$

where

- M_i denotes military capability,
- E_i denotes economic capability,
- T_i denotes technological capability.

Strategic equilibrium:

$$a_i^* = BR_i(a_{-i}).$$

12 Models in Warfare Economics and Military Science

Military planning employs predictive and optimization models.

Force attrition:

$$\frac{dR}{dt} = -\alpha B,$$

$$\frac{dB}{dt} = -\beta R.$$

Resource allocation:

$$\max \sum_t U_t$$

subject to logistics constraints.

Military doctrines are themselves models of conflict.

13 Models in Statistics

Statistical modelling seeks

$$P(Y|X).$$

Linear regression:

$$Y = X\beta + \epsilon.$$

Likelihood estimation:

$$L(\theta) = P(D|\theta).$$

Inference is therefore model-dependent.

14 Models in Computer Science

Algorithms are executable models.

Complexity:

$$T(n) = O(f(n)).$$

State machine:

$$s_{t+1} = f(s_t, a_t).$$

Computing systems operationalize abstract models.

15 Models in Artificial Intelligence

Machine learning models minimize loss.

$$\theta^* = \arg \min_{\theta} L(\theta).$$

Neural network:

$$h_{l+1} = \sigma(W_l h_l + b_l).$$

Transformer attention:

$$\text{Attention} = \text{softmax} \left(\frac{QK^T}{\sqrt{d}} \right) V.$$

16 Model Quality

Define quality

$$Q(M).$$

A general decomposition is

$$Q = \alpha E + \beta P + \gamma R + \delta S + \epsilon C,$$

where

- E = explanatory power,
- P = predictive power,
- R = robustness,
- S = simplicity,
- C = computational tractability.

17 The Trade-Off Theorem

Theorem 2. *No model can simultaneously maximize*

1. *realism,*
2. *tractability,*
3. *generality.*

Proof. Increasing realism introduces variables and constraints.

Increasing variables reduces tractability.

Increasing tractability requires simplification.

Simplification reduces realism.

Hence simultaneous maximization is generally impossible.

□

18 Model Maintenance

Let

$$\theta_t$$

denote model validity.

Assume

$$\frac{d\theta_t}{dt} = -\lambda\theta_t.$$

Then

$$\theta_t = \theta_0 e^{-\lambda t}.$$

Model maintenance seeks to minimize

$$\lambda.$$

19 Model Competition

Competing models

$$M_1, \dots, M_n$$

follow

$$\dot{s}_i = s_i(F_i - \bar{F}).$$

This equation governs the evolution of scientific paradigms, ideologies, institutions, and machine learning architectures.

20 Model Failure

Failures arise from

1. incorrect assumptions,
2. incorrect measurements,
3. incorrect dynamics,
4. structural breaks,
5. domain violations.

21 Model Deconstruction

Let

$$A_1, \dots, A_n$$

denote assumptions.

Outputs satisfy

$$C = f(A_1, \dots, A_n).$$

Sensitivity:

$$S_i = \frac{\partial C}{\partial A_i}.$$

Critical assumptions satisfy

$$|S_i| \gg 0.$$

22 Life Cycle of Models



Figure 1: Life cycle of models.

23 Meta-Theory of Models

Theorem 3 (General Model Theorem). *Every theory of models is itself a model.*

Proof. A theory of models possesses assumptions, definitions, state variables, logical rules, and conclusions.

Therefore it satisfies the definition of a model. □

Corollary 2. *No model theory can completely escape the limitations of modelling.*

24 Conclusion

Models are universal structures enabling finite agents to reason about complex realities. The construction, usage, maintenance, and deconstruction of models constitute a single dynamic process governing scientific progress, institutional evolution, technological development, strategic competition, and artificial intelligence. The resulting framework suggests that all disciplines may be understood as special cases of a general theory of modelling.

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Glossary

Model A structured representation of reality.

Ontology The entities assumed to exist.

State Variable A measurable quantity describing a system.

Parameter A numerical quantity governing model behaviour.

Prediction A forecast generated by a model.

Robustness Resistance to perturbations.

Structural Break A change in the underlying process.

Deconstruction Analysis of assumptions and hidden premises.

Meta-Model A model describing models.

Model Ecology A population of competing models.

The End